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NATIONAL INSTITUTE OF TECHNOLOGY GOA

Cuncolim, South Goa, Goa-403703

Programme Name: B.Tech

Minor Examination, October-2025

Course Name: Physics

Date: 07.10.2025

Duration: 1 Hour

Course Code: PH 100

Time: 10.00 am

Max. Marks: 20

ANSWER ALL QUESTIONS

1. X-rays with an energy of 300 keV undergo Compton scattering with a target. If the scattered X-rays are detected at 30° relative to the incident X-rays, determine the Compton shift at this angle, the energy of the scattered X-ray, and the energy of the recoiling electron. (3M)
2. (a) Show that at low temperatures the Planks formula for radiated energy $E_\lambda d\lambda = \frac{8\pi hc}{\lambda^5} \times \frac{1}{e^{hc/\lambda kT} - 1} d\lambda$ reduces to the Wein's law i.e. $E_\lambda d\lambda = \frac{8\pi hc}{\lambda^5} \times e^{-hc/\lambda kT} d\lambda$
(b) Show that at large temperatures the Planks formula for radiated energy $E_\lambda d\lambda = \frac{8\pi hc}{\lambda^5} \times \frac{1}{e^{hc/\lambda kT} - 1} d\lambda$ reduces to the Rayleigh-Jeans approximation i.e. $E_\lambda d\lambda = \frac{8\pi kT}{\lambda^4} d\lambda$ (4M)
3. Explain the wave nature of the electrons based on (1) Davisson-Germer Experiment and (2) G. P Thompson Experiment with the help of neat experimental diagrams (6M)
4. An electron has a speed of 500 m/s with an accuracy of 0.004%. Calculate the uncertainty with which we can locate the position of the electron. (Electron Mass $9.11 \times 10^{-31} \text{ kg}$) (3M)
5. Explain the particle nature of the photons with the help of experimental proof. (4M)

*** All the best ***



NATIONAL INSTITUTE OF TECHNOLOGY GOA

Cuncolim, South Goa, Goa-403703

Programme Name: B.Tech

Minor-II Examination, April-2026

Course Name: Engineering Physics

Date: 27.04.2026

Duration: 1 Hour

Course Code: PH100

Time: 10.00am

Max. Marks: 25

ANSWER ALL QUESTIONS

1. Describe the function of core and cladding in optical fiber. Differentiate between Step index and Graded index fiber. How the rays do propagates in Step index and Graded index fibers? (3M)
2. The normalized wave function for certain particle is $\Psi(x) = \sqrt{\frac{3}{\pi}} \cos x$, $-\frac{\pi}{2} < x < \frac{\pi}{2}$. Calculate the probability of finding the particle between $0 < x < \frac{\pi}{4}$ (5M)
3. Describe the construction and working of Fabry-Perot interferometer by oblique and normal incidence of light. (5 M)
4. Explain the following with help of the labeled diagrams with the interaction of matter and radiation in the process of (4 M)
 - (i) Induced absorption
 - (ii) Spontaneous Emission
 - (iii) Stimulated Emission.Which of the above process is maximized for LASER operations.
5. Explain the Lasing Action and Important characteristics of a LASER (3 M)
6. A monochromatic light of wavelength $\lambda = 600$ nm passes through a single slit, which has a width of 0.8 mm. What is the distance between the slit and the screen if the first minimum in the diffraction pattern is at a distance of 1.00 mm from the center of the screen? (Hint: Use small angle approximation, i.e., $\sin \theta \approx \tan \theta$) (5 M)

All the best



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NATIONAL INSTITUTE OF TECHNOLOGY GOA

Cuncolim, South Goa, Goa-403703

Programme Name: B.Tech

Minor Examination, February-2026

Course Name: Engineering Physics

Date: 13.02.2026

Duration: 1 Hour

Course Code: PH100

Time: 12.00-noon

Max. Marks: 25

ANSWER ALL QUESTIONS

1. Find the de Broglie wavelength of (i) an electron accelerated through a potential difference of 200 volts, and (ii) a 100 kg object moving with a speed 10 m/s. Comparing the results explain why the wave-like nature of matter is not observed every day for macroscopic objects? (4 M)
2. Explain the Davisson-Germer Experiment? Verify that the de Broglie waves were responsible for the findings of Davisson and Germer? (applied voltage 54 volts, angle 50° and nickel crystal spacing $d = 0.91 \text{ \AA}$) (6 M)
3. Show that the phase velocity of the de Broglie wave is greater than the velocity of light. Also, derive a relationship between group and phase velocity of matter waves (6 M)
4. Growing energy demand, fossil fuel depletion, and environmental concerns necessitate a transition in energy systems. Why is the transition to modern energy sources essential for sustainable national development? (2 M)
5. Explain and compare the energy conversion mechanisms of solar photovoltaic and solar thermal systems, and discuss their respective advantages. (4 M)
6. Rajasthan has benefited from Bhadla Solar Park project, one of the world's largest, boosting clean power generation and regional development. With reference to this, discuss the diverse applications of solar energy in modern energy systems. (3 M)

All the best



NATIONAL INSTITUTE OF TECHNOLOGY GOA

Cuncolim, South Goa, Goa-403703

Programme Name: B.Tech

Mid-Semester Examination, October-2025

Course Name: Engineering Physics

Date : 11.10.2025

Duration : 1 Hour 30 Minutes

Course Code : PH100

Time : 10.00 am

Max. Marks : 50

ANSWER ALL QUESTIONS

1. A measurement establishes the position of a proton with an accuracy of $\pm 1.00 \times 10^{-11} m$. Find the uncertainty in the proton's position 1.00s later. (Assume $v \ll c$, Proton mass = $1.672 \times 10^{-27} kg$) (5M)
2. State and explain Heisenberg's uncertainty principle and use it to (a) Prove the non-existence of electrons in the nucleus (If an electron is moving with Relativistic and Non-relativistic speeds) and existence of Protons, Neutrons and Alpha Particles inside the nucleus (b) Find the minimum energy of a simple harmonic oscillator. (10M)
3. An electron beam is accelerated from rest through a potential difference of 200 V. (5M)
 - (i) Calculate the associated wavelength.
 - (ii) This beam is passed through a diffraction grating of spacing $3 \text{ \AA}$. At what angle of deviation from the incident direction will be the first maximum observed?
4. Show that the phase velocity of the de Broglie wave is greater than the velocity of light. Also, derive a relationship between group and phase velocity of matter waves. (6M)
5. Discuss the following failures of classical mechanics and how does quantum mechanics overcome these failures. (6M)
 - (i) Stability of an Hydrogen Atom
 - (ii) Blackbody Radiation
 - (iii) Photoelectric Effect
6. An electron has a de Broglie wavelength of $2 \times 10^{-12} m$. Find its kinetic energy and the phase and group velocities of its de Broglie waves. ($m_0 = 9.1 \times 10^{-31} kg$) (5M)
7. The temperature of your skin is approximately $45^\circ C$. What is the wavelength at which the peak occurs in the radiation emitted from your skin? (Wien's displacement constant = $2.898 \times 10^{-3} mK$) (5M)
8. What is Compton effect? Show that the change in wavelength of the photon is given by $\Delta\lambda = \frac{h}{m_0 c} (1 - \cos\theta)$. In a Compton collision with an electron, a photon of violet light ($\lambda = 400 \text{ nm}$) is backward scattered through an angle 180° . (8M)
 - (a) How much energy is transferred to the electron in this collision?
 - (b) Compare the result with the energy, the electron would acquire in a photoelectric process with the same photon.

All the best



Roll No

25CSE1049

NATIONAL INSTITUTE OF TECHNOLOGY GOA

Cuncolim, South Goa, Goa-403703

Programme Name: B.Tech

Mid Semester Examination, March-2026

Course Name: Engineering Physics

Date: 02.03.2026

Duration: 1 Hour 30 Minutes

Course Code: PH100

Time: 10.00 am

Max. Marks: 50

ANSWER ALL QUESTIONS

1. What is the physical cause of Heisenberg's uncertainty principle and use it to find the minimum energy of the Harmonic Oscillator. (5 M)
2. A 50g of a golf ball moves at 60 m/s with a speed accuracy of 2%. What is the uncertainty in position? (3 M)
3. If the uncertainty in velocity is the same for a proton and an electron, what is the ratio of their position uncertainties? (Mass of the electron is $9.1 \times 10^{-31} \text{ kg}$ & Proton is $1.67 \times 10^{-27} \text{ kg}$) (4 M)
4. The lifetime of an excited state of an atom is $3 \times 10^{-3} \text{ s}$. What is the minimum uncertainty in its energy in eV? (4 M)
5. How Compton Effect differs from the Photoelectric Effect? Show that the change in wavelength of the photon is given by: $\lambda = \frac{h}{m_0 c} (1 - \cos \theta)$ (7 M)
6. What potential difference must be applied to stop the fastest photoelectrons emitted by a surface when electromagnetic radiation of frequency $1.5 \times 10^{15} \text{ Hz}$ is allowed to fall on it? The work function of the surface is 5 eV. (5 M)
7. (i) Discuss the shortcomings of classical physics (ii) what is Plank's quantum hypothesis to explain the observed spectrum of a Black Body? (6 M)
8. A small town is planning to replace LPG with an energy source generated from biodegradable waste.
 - a) Identify the energy source.
 - b) Explain how it is produced.
 - c) Justify its use by stating two advantages and two disadvantages. (6 M)
9. What is the physical significance of wave equation? Explain the Normalized and Orthogonal wavefunctions. (5 M)
10. A particle is at rest which can be represented by the following wave function: (5 M)

$$\psi(x, 0) = \begin{cases} \frac{Ax}{a}, & 0 \leq x \leq a \\ \frac{A(b-x)}{(b-a)}, & a \leq x \leq b \\ 0, & \text{otherwise} \end{cases}$$

where A, a and b are constants.

Normalize the given wavefunction by calculating A in terms of a and b.

All the best



NATIONAL INSTITUTE OF TECHNOLOGY GOA

Cuncolim, South Goa, Goa-403703

Programme Name: B.Tech

End Semester Examination, April-2026

Course Name: Engineering Physics

Date: 29.04.2026

Duration: 3 Hours

Course Code: PH100

Time: 10.00 am

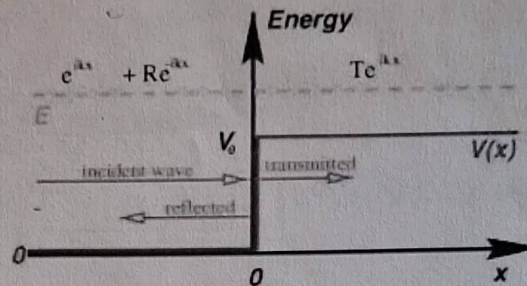
Max. Marks: 100

ANSWER ALL QUESTIONS

1. A particle of total energy E is moving in a one dimensional potential is given by

$$V(x) = 0 \text{ for } x < 0$$

$$V(x) = V_0 \text{ for } x > 0$$



Calculate the reflection and transmission coefficients for the case $E > V_0$ (6M)

2. If a beam of electrons impinge on any energy barrier of height 0.25 eV and of infinite width, find the fraction of electrons reflected and transmitted at the barrier if the energy of impinging electrons: (i) 0.30 eV (ii) 0.20 eV and (iii) 0.25 eV (4M)

3. An electron is trapped in a 1-D potential well of infinite height and a width of 0.1 nm. Calculate the energy required to excite it from its ground state to the fifth excited state. (5M)

4. A particle constrained to move along the x -axis in the domain $0 \leq x \leq L$ has a wave function $\psi(x) = \sin(n\pi x/L)$, where n is an integer. Normalize the wave function and evaluate the expectation value of its momentum. (5M)

5. Prove that the following wavefunction satisfies the wave equation (5 M)

$$\psi(x, t) = A_1 \cos k_1(x + vt) - A_2 \sin k_2(x - vt)$$

6. Find the condition under which the de Broglie wavelength equals the Compton wavelength. (Hint: $\lambda_C = \frac{h}{m_0 c}$ and $m = \frac{m_0}{\sqrt{1 - \frac{v^2}{c^2}}}$) (5 M)

7. A subatomic particle produced in a nuclear collision is found to have a mass such that $Mc^2 = 1228 \text{ MeV}$, with an uncertainty of $\pm 56 \text{ MeV}$. Estimate the lifetime of this state. Assuming that, when the particle is produced in the collision, it travels with a speed of 10^8 m/s , how far can it travel before it disintegrates? (5 M)

8. Derive the form of the Planck radiation formula in the limit case: (4 M)

(i) $\frac{hc}{\lambda k_B T} \ll 1$ (Large Wavelengths)

(ii) $\frac{hc}{\lambda k_B T} \gg 1$ (Small Wavelengths). Interpret the results

9. From a sodium surface, light of wavelength $3125 \text{ \AA}$ and $3650 \text{ \AA}$ causes emission of electrons whose maximum kinetic energy is 2.128 eV and 1.595 eV, respectively. Estimate Planck's constant and the work function of sodium. (5 M)

10. In Newton's ring experiment, the diameter of the 15th ring was found to be 0.59 cm, and that of the 5th ring was 0.336 cm. If the radius of the plano-convex lens is 100 cm, compute the wavelength of the light used. (5 M)
11. Discuss the phenomenon of obtaining an interference pattern due to thin films for an oblique incident of light (Transmitted Light). Prove the path difference between interfering waves (Cosine law: $2ut\cos r = n\lambda$). Write the expressions for maxima and minima conditions. (8 M)
12. Newton's rings are formed by the sodium light between glass plate and a convex lense are viewed normally. What will be the order of the dark ring, which will have double the diameter of that of 40th Dark ring. (5 M)
13. Calculate the acceptance angle and numerical aperture of the given optical fiber, having a spot diameter of 2.6 cm, when the distance between the screen and optical fiber is 3.0 cm. (5 M)
14. Explain the types of optical fibers based on mode and index profiles. Also explain the working principle of optical fiber. (5 M)
15. Explain the construction and working principle of the He-Ne Laser with the help of an energy level diagram. (5 M)
16. Explain the following pumping mechanism in a laser using the energy level diagrams (5 M)
 - (i) Two-Level Pumping
 - (ii) Three-Level Pumping
 - (iii) Four-Level Pumping
 Explain why two-level pumping is not feasible?
17. Explain the working principle of a boiling water reactor with the help of a schematic diagram. (5 M)
18. Consider a 100 cm² photovoltaic cell with reverse saturation current $I_0 = 10^{-12}$ A/cm². In full sun, it produces a short circuit current of 40 mA/cm² at 25 °C. Find the open circuit voltage at full sun and again for 50% sunlight. Plot the results. (5 M)
19. A coastal area has strong tidal movements, while a nearby inland region has a dam built across a flowing river. Suggest a suitable energy source for each location and compare these energy sources in terms of their working principle. (5 M)
20. Explain cell efficiency and fill factor with the help of an equation in solar cells. (3 M)

All the best



NATIONAL INSTITUTE OF TECHNOLOGY GOA

Programme Name: B.Tech

End Semester Examination, December-2025

Course Name: Physics

Date: 11.12.2025

Duration: 3 Hours

Course Code: PH100

Time: 10.00 am

Max. Marks: 100

ANSWER ALL QUESTIONS

1. The speed of an electron is measured to be $5.00 \times 10^3 \text{ m/s}$ to an accuracy of 0.003%. Find the minimum uncertainty in determining the position of this electron. (Electron Mass $9.11 \times 10^{-31} \text{ kg}$) (5 M)
2. Show that the phase velocity of the de Broglie wave is greater than the velocity of light. Also, derive a relationship between group and phase velocity of matter waves. (5 M)
3. If a beam of electrons impinges on any energy barrier (Tunnel) of height 0.70 eV and width of 10 nm, find the fraction of electrons transmitted through the tunnel if the energy of impinging electrons: (i) 0.30 eV and (ii) 0.40 eV (8 M)
4. A particle of mass m is in the state $\psi(x, t) = A e^{-ax^2}$, where A and a are positive real constants. (a) Find the value of A if this wavefunction is normalized? and (b) Calculate the expectation values of $\langle x \rangle$ and $\langle p \rangle$ (10M)
5. A parallel beam of light ($\lambda = 5890 \times 10^{-8} \text{ cm}$) is incident on a thin glass plate (refractive index=1.5) such that the angle of refraction into the plate is 60° . Calculate the smallest thickness of the glass plate which will appear dark by reflection. (5 M)
6. In Newton's ring experiment, the diameter of n^{th} and $(n+8)^{\text{th}}$ bright rings are 4.6 mm and 7.4 mm respectively. The radius of curvature of the lower surface of the lens is 2m. Determine the wavelength of light. (5M)
7. Discuss the phenomenon of obtaining interference pattern due to thin films for an oblique incident of reflected light. Prove the path difference between interfering waves (Cosine law: $2\mu t \cos r = n\lambda$). Write the expressions for maxima and minima conditions. (8M)
8. Find the ratio of populations of the two states in a He-Ne laser that produces light of wavelength 6328 Å at 27 °C. (Electron mass $m = 9.1 \times 10^{-31} \text{ kg}$, Planks constant $h = 6.62 \times 10^{-34} \text{ Js}$) (5 M)
9. Find the critical angles of the following (5 M)
 - (i) For an air-water boundary (The index of refraction of water is 1.33)
 - (ii) For an air-Glass boundary (The index of refraction of Glass is 1.48)
10. Calculate the minimum thickness of a soap-bubble film that results in constructive interference in the reflected light if the film is illuminated with light whose wavelength in free space is $\lambda = 600 \text{ nm}$. The index of refraction of the soap film is 1.33. (5 M)
11. Describe the function of core and cladding in optical fiber. Differentiate between Step index and Graded index fiber. How the rays do propagates in Step index and Graded index fibers? (5 M)

12. Find the peak wavelength of the blackbody radiation emitted by the sun, which has a surface temperature of approximately 5800 K. (Wein's constant = $2.898 \times 10^{-3} \text{ m.K}$) (5 M)
13. Explain the different types of solar cell structures and how they are influencing efficiency, applications, and overall performance (6 M)
14. What is the physical significance of Schrödinger wave equation? Explain the Normalized and Orthogonal wave functions. Derive time-dependent Schrodinger equation in three dimensions for a non-relativistic free particle. (8 M)
15. What is nuclear reactor and explain its components. Explain the construction and working of pressurized water reactor (5 M)
16. Consider a p-n junction diode at 25°C with a reverse saturation current of 10^{-9} A . Find voltage drop across the diode when it is carrying 1A current. (5 M)
17. At a certain time, the normalized wave function of a particle moving along the X-axis has the form given by:
- $$\psi(x) = \begin{cases} x + \beta & \text{for } -\beta < x < 0 \\ -x + \beta & \text{for } 0 < x < \beta \end{cases} \text{ and zero elsewhere}$$
- Find the value of β and the probability that the particle positions between $x = -\frac{\beta}{2}$ and $x = \beta$ (5 M)

All the best